HW #1

1) Speed of Light : 3.00×10^8 m/s

① s → year : year $\xrightarrow{365}$ day $\xrightarrow{24}$ hour $\xrightarrow{60}$ minute $\xrightarrow{60}$ second

= year $\xrightarrow{31\,536\,000}$ second

m → km = km $\xrightarrow{1000}$ m

② ~~$3.00 \times 10^8 \frac{1m \cdot 1000}{1s \cdot 31536000} = 9.512.94$ km/year~~

$3.00 \times 10^8 \frac{1/1000 \text{ km}}{1/31536000 \text{ year}} = 9.51 \times 10^{12}$ km/year

2)

a) $3.00 \times 10^8 \frac{m}{s} \cdot \frac{100 \text{ cm}}{10^9 \text{ ns}} \cdot \frac{1/2.52 \text{ in}}{1} \cdot \frac{1/12 \text{ ft}}{1} = 9.92 \times 10^{-1}$ ft/ns

b) $9.92 \times 10^{-1} \frac{ft}{ns} \cdot \frac{1/5280 \text{ mile}}{1/10^9 s} = 1.88 \times 10^5 \frac{\text{mile}}{s}$

c) $3.00 \times 10^8 \frac{m}{s} \cdot \frac{1/1.5 \times 10^{11} m}{1/3600 h} = \cancel{1.25 \times 10^{16}} \frac{AU}{h}$

$7.2 \frac{AU}{h}$

3)

$4.5 L \doteq 4.5 \times 0.001 m^3 = 0.0045 m^3$

$4.5 \times 10^{-3} m^3 / 22 m^2 = 2.05 \times 10^{-4} m = 2.05$ mm

4)

a) $100 \text{ yard} \cdot 36 \text{in} \cdot 2.54 \text{cm} \cdot \frac{1}{100} m = 91.44 m$

b) $(\sqrt{12.5 \text{acre}} \cdot 484 \cdot \text{yd} \cdot 36 \text{in} \cdot 2.54 \text{cm} \cdot \frac{1}{100} m)^2 \cdot \frac{1}{10000} =$

44 817.57 hectare

5)

$x = l \quad a = 1/t^2 \quad t = t$

$l = (1/t^2)^m t^n$

$$l = l^m / t^{2m} \cdot t^n$$

$$\cancel{n=1} \quad l = l^m t^{n-2m}$$

$$\cancel{l=} \quad 0 = l^{m-1} t^{n-2m}$$

$$m = 1 \quad n = 2$$

6)

$$90.0\ km/h \cdot 5.5h / 100\ km/h = 4.95h$$
$$= 4h\ 57\ min$$

$$5.5h - 4.95h = .55h$$
$$5h\ 30\ min - 4h\ 57\ min = 33\ min$$

7)

a) $11.0\ m/s \cdot 1.00\ min \cdot 60\ s = 660\ m$

$+ 16.0\ m/s \cdot 2.00\ min \cdot 60\ s = 1920\ m$

$2580\ m / 180\ s = 14.3\ m/s$

b) $(660\ m - 1920\ m) / 180\ s = -7\ m/s$

8)

$10\ km / (6\ km/h + 4\ km/h) = 1h$

$1h \cdot 20\ km/h = 20\ km$